



**UNIVERSITÀ
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Image Denoising with Autoencoders

Reconstruction Quality, Robustness and Latent-Space Analysis

Course
Deep Learning

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Image denoising problem

A clean image x is corrupted by a random process:

$$\tilde{x} = \mathcal{C}(x, \varepsilon).$$

A denoising model f_θ produces:

$$\hat{x} = f_\theta(\tilde{x}).$$

The objective is to make $\hat{x}$ close to x , while avoiding:

- ▶ excessive smoothing;
- ▶ loss of edges and class-specific details;
- ▶ sensitivity to small changes in the corruption process.

Three levels of evaluation

- ① Pixel fidelity
- ② Robustness to corruption shift
- ③ Latent-space recovery

Research questions and contributions

- ① How do classical filtering, a fully connected autoencoder and a convolutional autoencoder compare at the training noise level?
- ② How robust are the trained models to weaker and stronger Gaussian noise?
- ③ Do models trained on Gaussian noise generalize to salt-and-pepper noise and random masking?
- ④ Does reconstruction move corrupted samples back toward their clean latent representations?
- ⑤ Are semantic classes and local neighborhoods preserved in latent space?

Dataset: Fashion-MNIST



- ▶ 60,000 training images
- ▶ 10,000 test images
- ▶ Grayscale, 28×28 pixels
- ▶ Tensor shape: $[1, 28, 28]$
- ▶ Pixel range: $[0, 1]$
- ▶ Ten clothing categories

Use of labels

Class labels are not used to train the autoencoders. They are used only for the final semantic analysis of the latent spaces.

Corruption processes

Gaussian noise

$$\tilde{x} = \text{clip}(x + \varepsilon, 0, 1), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I).$$

Training condition: $\sigma_{\text{train}} = 0.3$.

Salt-and-pepper noise:

A random fraction of pixels is replaced by 0 or 1.

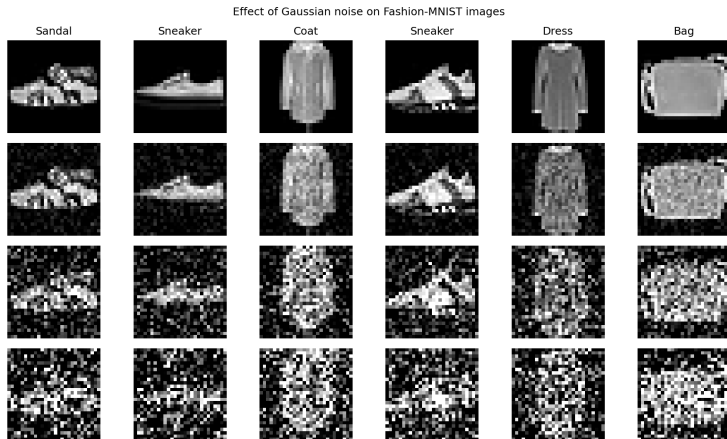
Test probability: $p = 0.2$

Random masking:

A random fraction of pixels is removed by setting it to zero.

Test probability: $p = 0.3$

Effect of Gaussian noise intensity



Key observation

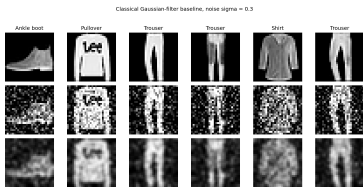
Increasing σ progressively hides local texture and object silhouettes; clipping keeps all corrupted pixels in $[0, 1]$.

Classical baseline: Gaussian filtering

A fixed 3×3 kernel is applied independently to each image:

$$K = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

- ▶ No trainable parameters
- ▶ Reduces local random fluctuations
- ▶ Introduces blur at edges and fine details



Evaluation metrics

Mean squared error

$$\text{MSE} = \frac{1}{CHW} \sum_{c,i,j} (\hat{x}_{cij} - x_{cij})^2.$$

Lower values are better.

Why both?

MSE is the optimization objective, while PSNR expresses the same reconstruction error on a logarithmic scale commonly used in image restoration.

Peak signal-to-noise ratio

Since the maximum pixel value is 1:

$$\text{PSNR} = 10 \log_{10} \left(\frac{1}{\text{MSE}} \right).$$

Higher values are better.

What does an MSE denoiser learn?

For a fixed corrupted observation $\tilde{x}$, consider the conditional risk:

$$R(a \mid \tilde{x}) = \mathbb{E} \left[\|X - a\|_2^2 \mid \tilde{X} = \tilde{x} \right].$$

Expanding the square:

$$R(a \mid \tilde{x}) = \mathbb{E}[\|X\|_2^2 \mid \tilde{x}] - 2a^\top \mathbb{E}[X \mid \tilde{x}] + \|a\|_2^2.$$

Therefore:

$$\nabla_a R = 2a - 2\mathbb{E}[X \mid \tilde{x}] = 0,$$

and the optimal estimator is

$$f^*(\tilde{x}) = \mathbb{E}[X \mid \tilde{X} = \tilde{x}].$$

Consequence

The optimal prediction depends on the corruption distribution. A Gaussian-noise denoiser is not automatically optimal for masking or other unseen corruptions.

Fully connected denoising autoencoder

Input $1 \times 28 \times 28$
↓ Flatten
 $784 \rightarrow 256 \rightarrow 64 \rightarrow 16$
↓ latent representation
 $16 \rightarrow 64 \rightarrow 256 \rightarrow 784$
↓ Unflatten
Output $1 \times 28 \times 28$

- ▶ ReLU hidden activations
- ▶ Sigmoid output
- ▶ Latent dimension: 16
- ▶ Parameters: 437,664
- ▶ Every neuron can use global image information

Limitation

Flattening removes the explicit two-dimensional organization of the pixels.

Convolutional denoising autoencoder

Input $1 \times 28 \times 28$

↓ Conv2d $1 \rightarrow 16$, stride 2
 $16 \times 14 \times 14$

↓ Conv2d $16 \rightarrow 32$, stride 2
 $32 \times 7 \times 7$

↓ Linear $1568 \rightarrow 16$
Latent vector $z \in \mathbb{R}^{16}$

↓ Linear $16 \rightarrow 1568$
 $32 \times 7 \times 7$

↓ ConvTranspose2d $32 \rightarrow 16 \rightarrow 1$
Output $1 \times 28 \times 28$

Inductive bias

- ▶ local receptive fields;
- ▶ shared filters;
- ▶ spatial feature maps;
- ▶ parameter efficiency.

Model size

Latent dimension: 16.

Trainable parameters: 65,025.

Architectural inductive biases

Fully connected model

- ▶ Global connections
- ▶ No explicit translation-related prior
- ▶ High parameter count
- ▶ Potential advantage when reconstruction requires global dependencies

$$\frac{65,025}{437,664} \approx 14.9\%.$$

The convolutional model uses only 14.9% of the FC-AE parameters.

Convolutional model

- ▶ Local receptive fields
- ▶ Shared filters
- ▶ Spatial feature hierarchy
- ▶ Strong parameter efficiency

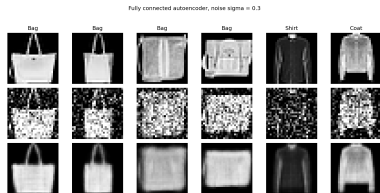
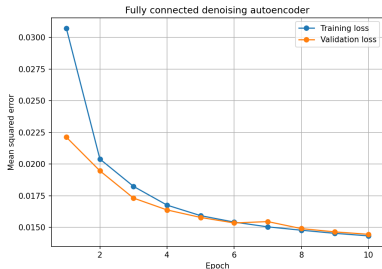
Training protocol

- ▶ Training set: 54,000
- ▶ Validation set: 6,000
- ▶ Test set: 10,000
- ▶ Batch size: 32
- ▶ Epochs: 10
- ▶ Latent dimension: 16
- ▶ Optimizer: Adam
- ▶ Learning rate: 10^{-3}
- ▶ Loss: MSE
- ▶ Training noise: $\sigma = 0.3$
- ▶ Best checkpoint selected by validation loss
- ▶ Fixed random seeds for reproducibility

Training pair

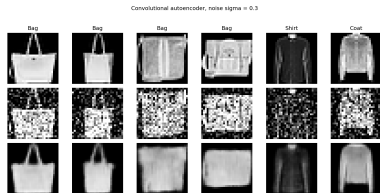
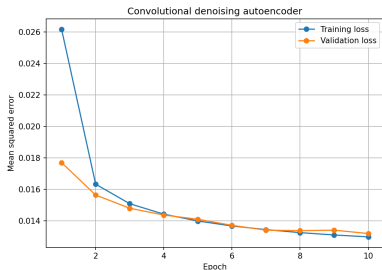
The input is the corrupted image $\tilde{x}$, while the target remains the corresponding clean image x .

Fully connected autoencoder: training



- ▶ Best validation MSE: approximately 0.01446.
- ▶ The model removes most visible noise.
- ▶ Reconstructions are smoother than the clean targets.

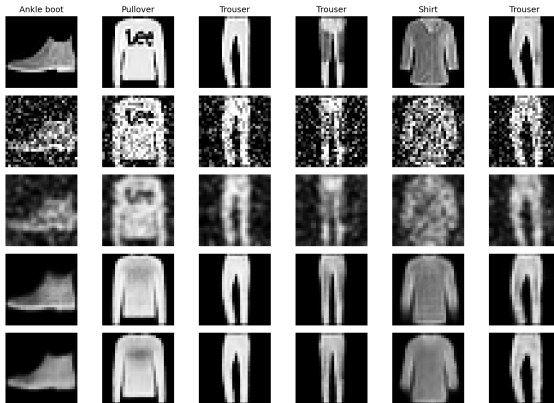
Convolutional autoencoder: training



- ▶ Best validation MSE: approximately 0.01319.
- ▶ Edges and local shapes are slightly better defined than with the FC-AE.
- ▶ No obvious checkerboard artifacts were observed.

Qualitative comparison at $\sigma = 0.3$

Denoising comparison on Fashion-MNIST, noise sigma = 0.3



Visual summary

Gaussian filtering blurs the image, while both autoencoders recover clearer silhouettes; the Conv-AE generally preserves local contours better.

Test results at the training noise level

Method	MSE ↓	PSNR ↑
Noisy input	0.052624	12.809 dB
Gaussian filter	0.025950	15.922 dB
FC autoencoder	0.014341	19.141 dB
Conv autoencoder	0.013170	19.512 dB

Main result

The Conv-AE improves MSE by approximately

$$\frac{0.014341 - 0.013170}{0.014341} \times 100 \approx 8.16\%$$

over the FC-AE, while using far fewer parameters.

Robustness to Gaussian noise intensity

Both autoencoders are trained only at

$$\sigma_{\text{train}} = 0.3$$

and evaluated at

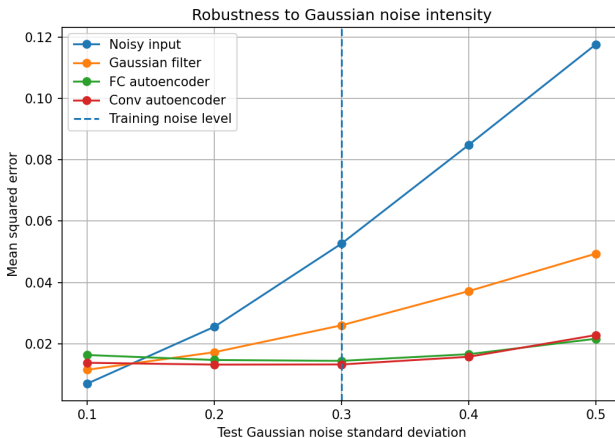
$$\sigma_{\text{test}} \in \{0.1, 0.2, 0.3, 0.4, 0.5\}.$$

Questions

- ▶ Is a learned denoiser still useful when the test image is cleaner?
- ▶ How quickly does performance degrade under stronger noise?
- ▶ Does the convolutional advantage persist outside the training condition?

All methods are evaluated on the complete test set using identical corrupted inputs.

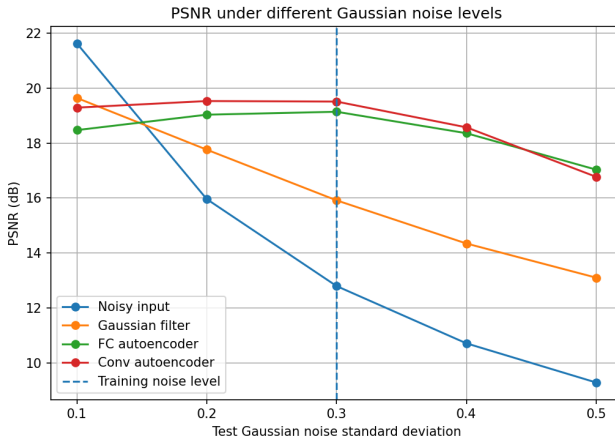
Robustness results: MSE



Main reading

The learned denoisers are strongest near $\sigma = 0.2$ – 0.3 . At $\sigma = 0.1$, denoising can worsen already-clean inputs; at $\sigma = 0.5$, the FC-AE slightly outperforms the Conv-AE.

Robustness results: PSNR



Main reading

PSNR confirms the MSE trend: the best denoiser depends on the mismatch between training and test corruption, not only on model class.

Interpreting the robustness experiment

Weak noise: reconstruction bias

With $\sigma = 0.1$, the input is already close to the clean target. Passing it through a model trained on stronger corruption can remove useful detail and move the image away from the target.

Intermediate noise

Around the training condition, both autoencoders provide substantial gains, and the convolutional inductive bias is most effective.

Severe noise

When information is heavily destroyed, local features become less reliable. The Conv-AE advantage shrinks and reverses at $\sigma = 0.5$.

Generalization to unseen corruptions

The models are trained only with Gaussian noise, then tested without retraining on:

Salt-and-pepper

$$p = 0.2$$

Local impulsive corruption.
Pixel values are replaced by 0 or 1.

Distribution shift

The marginal appearance of corruption and the conditional distribution $p(x | \tilde{x})$ differ from the training condition.

Random masking

$$p = 0.3$$

Missing-data corruption. Pixel values are removed by setting them to zero.

Results on unseen corruption mechanisms

Salt-and-pepper	MSE	PSNR
Noisy input	0.083863	10.803
Gaussian filter	0.028941	15.467
FC-AE	0.015152	18.852
Conv-AE	0.014507	18.985

Random masking	MSE	PSNR
Noisy input	0.061975	12.965
Gaussian filter	0.035255	15.117
FC-AE	0.031592	15.588
Conv-AE	0.036708	14.703

Interpretation

Both autoencoders transfer well to salt-and-pepper noise. Random masking is harder: the FC-AE is best, while the Conv-AE is slightly worse than the Gaussian filter.

Why does random masking behave differently?

Gaussian and impulsive noise

Information is perturbed locally, but much of the original spatial support remains visible.

Random masking

Information is removed.

Reconstruction requires inferring missing regions from broader dependencies.

- ▶ The Conv-AE local inductive bias is highly effective near its training distribution.
- ▶ The FC-AE global connectivity may help when larger-scale relationships are required.
- ▶ This is a plausible interpretation of the results, not a causal proof.

General lesson

Architectural priors create both advantages and failure modes.

From pixel reconstruction to latent recovery

For each test image:

$$z_{\text{clean}} = f_{\text{enc}}(x), \quad z_{\text{noisy}} = f_{\text{enc}}(\tilde{x}), \quad z_{\text{reconstructed}} = f_{\text{enc}}(\hat{x}).$$

The analysis considers:

- 1 Euclidean distance from the clean latent code

$$\|z_{\text{clean}} - z_{\text{noisy}}\|_2, \quad \|z_{\text{clean}} - z_{\text{reconstructed}}\|_2;$$

- 2 two-dimensional PCA visualization;
- 3 k -nearest-neighbor semantic accuracy;
- 4 preservation of local neighborhoods.

Important caution

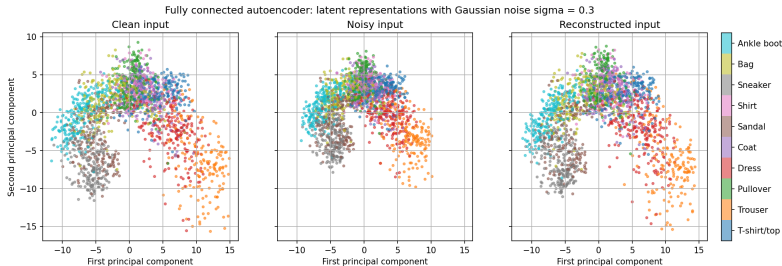
PCA is a two-dimensional visualization of a 16-dimensional space.
Quantitative conclusions are computed in the original latent space.

Pointwise latent recovery

Metric	FC-AE	Conv-AE
Mean clean-noisy distance	3.7749	2.9664
Mean clean-reconstructed distance	1.6587	2.1491
Relative latent recovery	56.06%	27.55%
Samples with improved distance	98.25%	82.07%

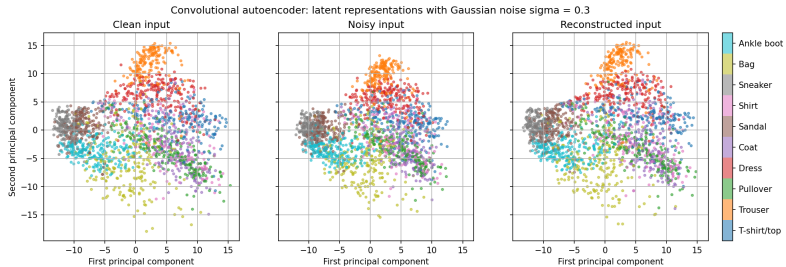
- ▶ The Conv-AE latent code is less perturbed by Gaussian noise initially.
- ▶ The FC-AE reconstruction recovers a larger fraction of its latent displacement.
- ▶ Absolute distances must not be compared directly across models because the latent coordinate systems can have different scales.

FC-AE latent representations: PCA



- In the PCA projection, noise shifts the latent representations and makes several class regions more overlapping.
- Reconstructed samples move back toward the clean regions.
- Semantically similar classes remain strongly overlapping.

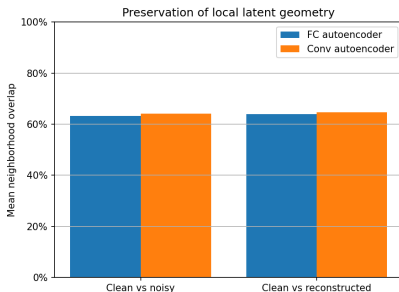
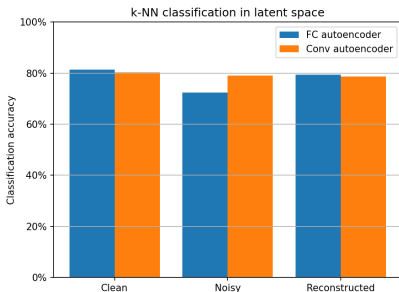
Conv-AE latent representations: PCA



- Clean and noisy projections are more similar than for the FC-AE.
- Reconstruction changes the representation less dramatically.
- Visual overlap in PCA does not imply identical geometry in 16 dimensions.

Semantic accuracy/Neighborhood preservation

Quantitative analysis of latent representations



k-NN accuracy

FC-AE recovers semantic accuracy after denoising; Conv-AE remains almost unchanged:

FC-AE: 72.45% $\rightarrow$ 79.35%.

Conv-AE: 78.75% $\rightarrow$ 78.55%.

Neighborhood overlap

Local geometry changes little after reconstruction:

FC-AE 63.20% $\rightarrow$ 64.00%

Conv-AE 64.48% $\rightarrow$ 64.48%.

Main conclusions

- ① Both autoencoders strongly outperform the noisy input and Gaussian filter at the training corruption level.
- ② The Conv-AE gives the best standard reconstruction with only 14.9% of the FC-AE parameters.
- ③ This advantage is not universal: it shrinks under severe Gaussian noise and reverses under random masking.
- ④ Better pixel reconstruction does not necessarily imply better semantic or geometric recovery in latent space.
- ⑤ The Conv-AE latent representation is more stable to noise, while the FC-AE shows stronger recovery after reconstruction.
- ⑥ Neither model exactly restores the original local latent geometry.

Limitations

Limitations

- ▶ Fashion-MNIST only (simple dataset)
- ▶ Fixed latent dimension
- ▶ Particular choice of hyperparameters
- ▶ Synthetic corruptions
- ▶ MSE can favor smooth predictions
- ▶ PCA is only a low-dimensional projection

Possible extensions

- ▶ Mixed-corruption training
- ▶ Different latent dimensions
- ▶ Structural or perceptual losses
- ▶ Natural-image datasets
- ▶ Intrinsic-dimension analysis

Thank you for your attention.



Appendix: Gaussian robustness table I

σ	Method	MSE	PSNR
0.1	Noisy input	0.006939	21.619
	Gaussian filter	0.011443	19.638
	FC-AE	0.016258	18.469
	Conv-AE	0.013717	19.286
0.2	Noisy input	0.025504	15.964
	Gaussian filter	0.017187	17.759
	FC-AE	0.014626	19.030
	Conv-AE	0.013125	19.525
0.3	Noisy input	0.052680	12.804
	Gaussian filter	0.025973	15.918
	FC-AE	0.014346	19.136
	Conv-AE	0.013178	19.506

Appendix: Gaussian robustness table II

σ	Method	MSE	PSNR
0.4	Noisy input	0.084907	10.723
	Gaussian filter	0.037130	14.350
	FC-AE	0.016531	18.356
	Conv-AE	0.015682	18.571
0.5	Noisy input	0.117736	9.299
	Gaussian filter	0.049381	13.109
	FC-AE	0.021573	17.032
	Conv-AE	0.022770	16.771

The FC-AE is slightly better at the most severe Gaussian corruption, even though the Conv-AE is better near the training condition.

Appendix: latent-space metrics

1. Pointwise latent recovery

For each test image, we compare how far the noisy and reconstructed latent codes are from the clean code:

$$R_i = 1 - \frac{\|z_{\text{clean}}^{(i)} - z_{\text{reconstructed}}^{(i)}\|_2}{\|z_{\text{clean}}^{(i)} - z_{\text{noisy}}^{(i)}\|_2}.$$

2. k -NN semantic accuracy

Each query code is assigned the majority label among its $k = 5$ nearest **clean training codes**.

Appendix: latent-space metric II

3. Local-neighborhood preservation

For $k = 10$, we compare the clean neighbors of sample i with its neighbors after corruption or reconstruction:

$$O_i = \frac{|\mathcal{N}_k^{\text{clean}}(i) \cap \mathcal{N}_k^{\text{comparison}}(i)|}{k}.$$

$O_i = 1$ means that all local neighbors are preserved; $O_i = 0$ means that none are preserved.